

Plexus — Agent Mesh Protocol v1.3

A protocol for autonomous agents to dispatch work to each other over MQTT — across laptops, VPNs and containers, none of which can accept an inbound connection. Broker root: `agents` .

Delivery is **push end to end**. Nothing on a delivery path polls: the broker pushes over a persistent MQTT session, the host pushes the job into an executor the moment it arrives, results are pushed back at QoS 1 (retained), and the web panel receives Server-Sent Events. The only periodic timers left are a supervisory watchdog and a slow filesystem reconciler, neither of which carries a message.

Implementations

Three, sharing no code. This document is what they agree on:

	Language	What it is
<code>src/</code>	TypeScript	Host plugin for OpenClaw
<code>packages/agent</code>	JavaScript	<code>plexus-agent</code> — a client library and plugin host
<code>hosts/hermes/</code>	Python	Host plugin for Hermes Agent

A specification with one implementation is a description of that implementation. The Python and JavaScript ones were written independently against this document, and CI stands them up against a single broker on every push to check that each can discover, delegate to and answer the other with lineage intact — see `hosts/hermes/tests/test_interop.py` .

Conformance

An implementation is conformant when all of the following hold. Each is directly observable from another client on the same broker, and each has a test:

1. A retained profile appears at `<root>/registry/<agentId>/profile` on connect, and is **deleted** (empty retained payload) on clean shutdown.
2. Presence is published as a **last will**, not by a heartbeat.
3. The client id is **stable across restarts**, so an invoke published during downtime is queued and delivered on reconnect.
4. Every job reaches a **terminal result**, retained, on every path — including rejection, unknown service, duplicate id, hop-limit refusal, timeout, cancellation, and an executor that throws.
5. Results are retained; **events are not**.
6. Job topics are **always owner-scoped**; there is no unscoped form.
7. A delegated ask carries `parentJobId` , `rootJobId` and `depth` , and a request arriving deeper than `maxDepth` is refused **before** any work starts.
8. `${VAR}` expands **before** `{{args}}` wherever prompts are rendered.

Why MQTT

An agent is not a service. It is intermittent, slow, context-bound to a machine, and liable to stop existing halfway through a thought. Treating it as an HTTP endpoint means rebuilding, in application code, every mechanism that already exists in a transport designed for exactly this shape.

The agent's reality	What the transport already provides
Offline for hours at a time	Persistent session: work is queued and delivered on reconnect
Unknown to its peers	Retained capability profile — discovery with no announcement round
Thinks for minutes	Streamed milestones, terminal result retained for late collection
Dies mid-task	Last will: presence published by the broker , not by a heartbeat service
One of many requesters	Per-owner isolation as a subscription filter , enforceable in broker ACLs
Behind NAT or a VPN	One outbound connection; nothing exposed

Every guarantee below is a property of the transport, not of this implementation. That is what makes the protocol small enough to implement in an afternoon — see [examples/minimal-agent.mjs](#).

Delivery guarantees

Stated exactly, because "reliable" means nothing on its own.

Message	Delivery	Retained	Notes
<code>invoke</code>	at-least-once	no	QoS 1. A redelivery after reconnect can duplicate a job; the bridge rejects a <code>jobId</code> already active
<code>events</code>	at-least-once	no	A subscriber joining mid-job sees no earlier milestones. Progress is advisory, never a source of truth
<code>result</code>	at-least-once	yes	Last write wins. Readable indefinitely by clients that connect afterwards
<code>profile / status</code>	at-least-once	yes	Empty payload deletes the retained message and means the agent has left

Ordering holds per topic only. `events` and `result` are different topics: a terminal result may be observed before a milestone published earlier. Never infer sequence across topics — use the `ts` field.

Durability covers the window in which an agent is disconnected, and depends entirely on the client id being stable. With a changing id every reconnect is a new session, and jobs published during the gap are lost with the old one.

Exactly-once is not offered. MQTT QoS 2 would give it hop-by-hop and still not end-to-end, because an executor can complete work and die before publishing. Assume at-least-once and make capabilities idempotent.

Job lifecycle

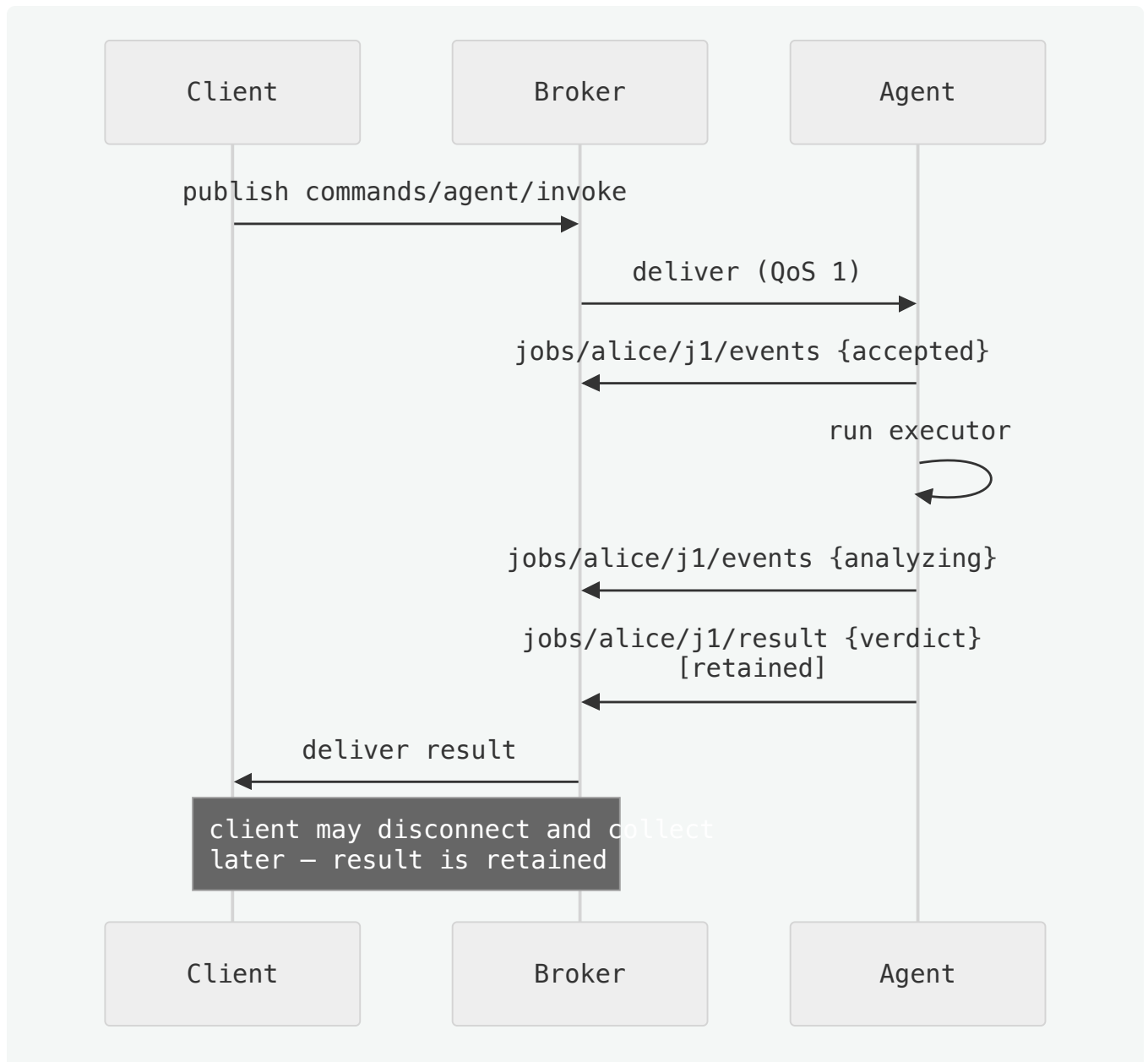


Terminal states are `done`, `error`, `timeout`, `cancelled`, `rejected`, `duplicate`. **Every path reaches one**, and each publishes a retained result — a client is never left waiting on a job that quietly stopped existing.

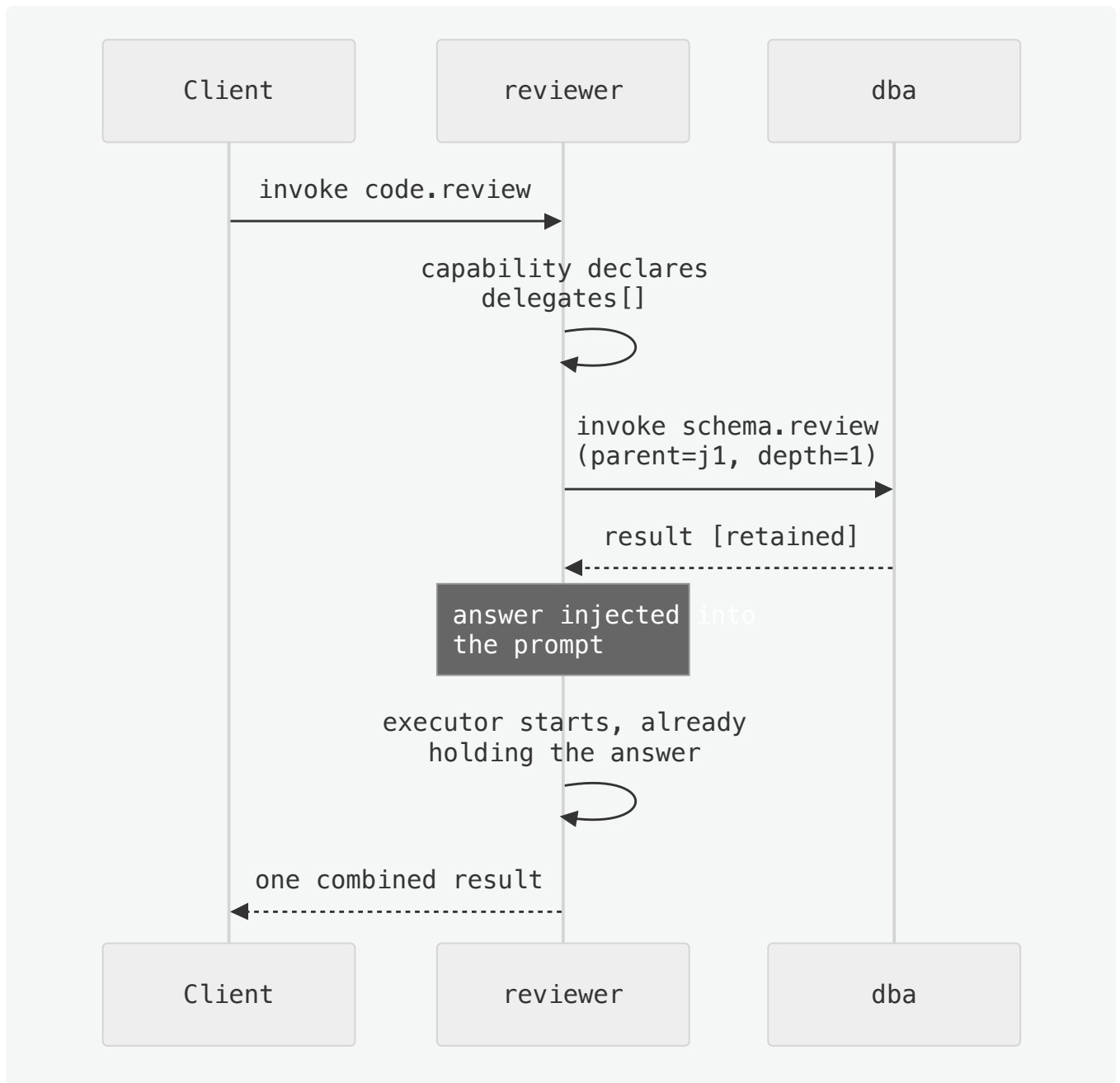
`started` may be re-entered once per watchdog re-dispatch, bounded by `maxDepth`-independent retry limits. Re-entry is visible as a `requeued` event on the timeline.

Sequence

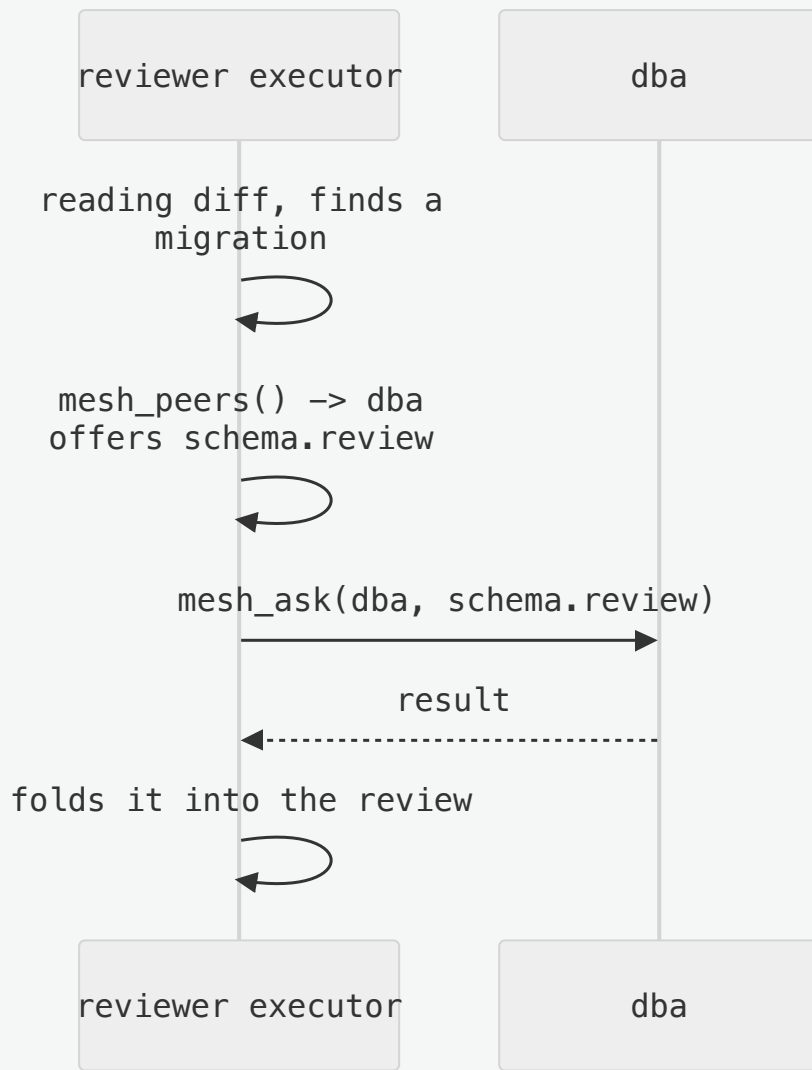
A directed job:



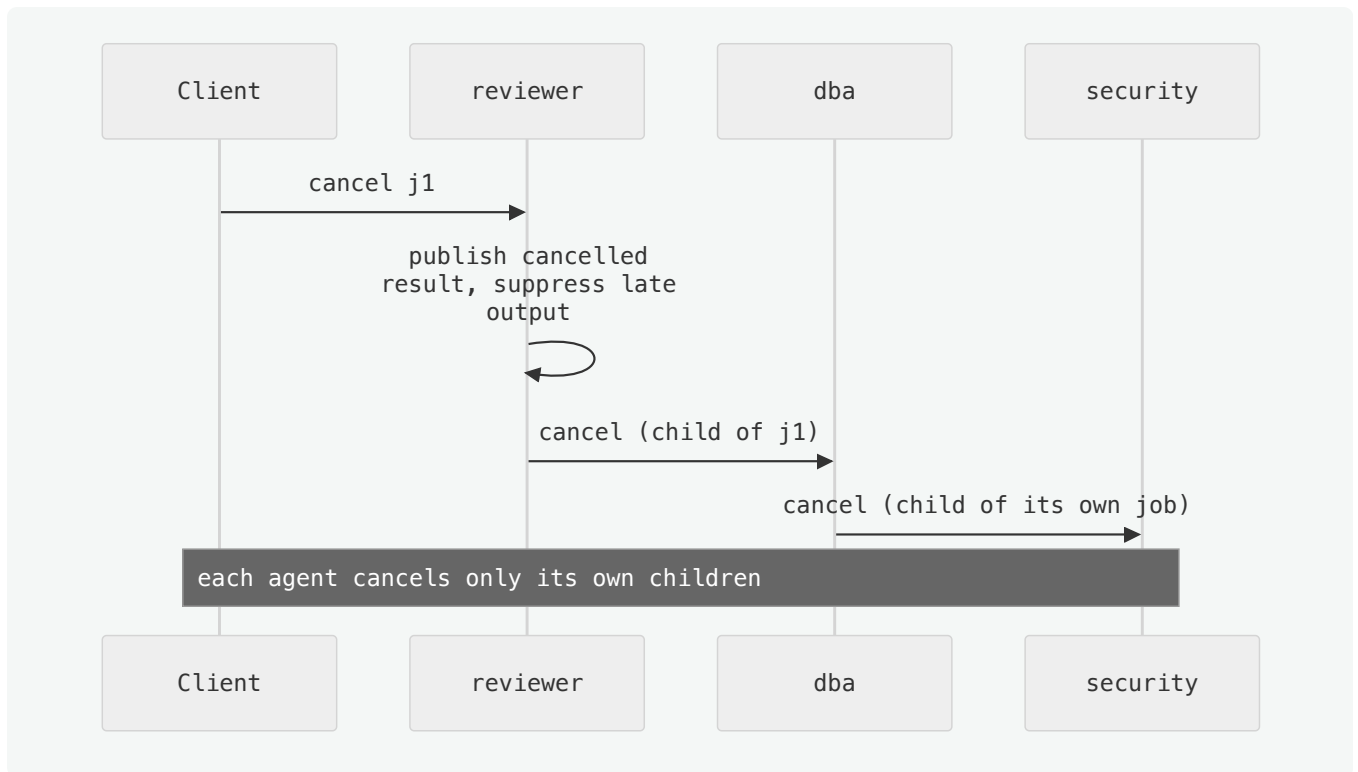
Declared delegation — the bridge gathers dependencies *before* the executor starts:



Dynamic delegation — the executor discovers the need mid-job:



Cancel propagates down a chain without any agent knowing its shape:



Failure modes

What actually happens, rather than what is hoped for.

Situation	Behaviour
Broker unreachable	Agent retries every 5s. Jobs published meanwhile are queued by the broker and delivered on reconnect, provided the client id is stable
Agent offline when a job is published	Queued in its persistent session, delivered on reconnect. This is the main reason <code>clean: false</code> matters
Agent restarts mid-job	In-memory job state is lost; the executor's own retained result still lands. The watchdog does not survive the restart, so a job whose executor also died goes unresolved
Executor finishes without publishing	Watchdog re-dispatches after the silence window, up to twice, then publishes a terminal <code>error</code> . Re-dispatch can duplicate work if the capability is not idempotent
Executor never terminates	<code>maxJobDurationMs</code> publishes a terminal error regardless of liveness
Peer never answers a delegated ask	<code>askTimeoutMs</code> fails the ask. Declared: optional dependency degrades, required fails the job before the executor starts
Delegation cycle (A→B→A)	<code>maxDepth</code> refuses before publishing — a cycle allowed to start would run until something else stopped it
Duplicate <code>jobId</code>	Rejected with a <code>duplicate</code> result if already active. Note ids are not namespaced by asker, so two agents <i>can</i> collide
Unknown service	Rejected with a terminal <code>error</code> naming the service
Missing <code>requestedBy</code>	Rejected. The error goes to <code>jobs/public/<id>/result</code> , not the sender's scope — a client that omitted it will not see the rejection on its own filter
Result exceeds broker message limit	Publish fails at the broker. No chunking exists; large payloads should be published elsewhere and referenced
Two agents share a client id	Broker kicks each in turn. Detected at 5+ connects in 60s; the bridge takes a distinct id and reports degraded durability

Registry (retained)

Topic	Direction	Payload
<code>agents/registry/<agentId>/profile</code>	agent → all (retained)	<code>{ agentId, displayName, status, protocolVersion, capabilities[], ownerPolicy }</code>
<code>agents/registry/<agentId>/status</code>	agent → all (retained + LWT)	<code>{ status: online offline }</code>

`ownerPolicy` is `{ required: bool, verified: bool }` — it tells you what this deployment actually enforces, so you never have to infer enforcement from the version number.

Commands

Topic	Direction	Notes
<code>agents/commands/<agentId>/invoke</code>	you → agent	<code>{ service, args, requestedBy, jobId? }</code> — requestedBy is REQUIRED and is now enforced
<code>agents/commands/<agentId>/query</code>	you → agent	<code>{}</code> lists services; <code>{ jobId }</code> checks state. Reply on <code>.../query/reply</code>
<code>agents/commands/<agentId>/cancel</code>	you → agent	<code>{ jobId, requestedBy }</code> . Reply as a terminal <code>cancelled</code> result
<code>agents/commands/<agentId>/config</code>	you → agent	service CRUD. Reply on <code>.../config/reply</code>

Capabilities are data, not code

The bridge is **capability-agnostic**. It contains no service name — not `code.review`, not anything else. A capability is a JSON entry in `services.json`:

```
{ "service": "code.review",
  "description": "...",
  "requestSchema": { "repo": "string (owner/name)", "pr": "number" },
  "handler": "session",
  "prompt": "Review pull request {{pr}} in {{repo}}. ..." }
```

`invoke` names a `service`; the bridge looks it up, renders its `prompt` with `{{jobId}}`, `{{requestedBy}}` and each key of `args`, and hands the result to an executor. That is the entire coupling. Everything domain-specific lives in the prompt template.

Capabilities can be added, updated and removed at runtime over the `config` command (or the panel) with no restart and no code change — the retained profile republishes automatically. The catalog shipped here is an **example**; replace it wholesale for a different domain and the transport is unchanged.

Deployment values: `${ENV_VAR}` in prompts

A capability definition should stay portable, but real prompts need deployment-specific values — Slack ids, channel names, internal repo paths. Reference them as `${VAR}`:

```
"prompt": "Review PR {{pr}} in {{repo}}. DM the summary to ${SLACK_REVIEW_RECIPIENTS}."
```

and supply them per deployment, either in plugin config:

```
"mesh": {  
  "promptVars": { "SLACK_REVIEW_RECIPIENTS": "U07XXXXXX,U08YYYYYYY" }  
}
```

or from the environment. **Config is checked first**, then `process.env`. Prefer config for identifiers — it is versioned with the rest of your deployment and survives any regeneration of a service-managed environment file. Keep genuinely secret values in the environment.

The point of the split: one `services.json` runs everywhere, and only the variable bindings change. Without it, every deployment forks the catalog and every upstream catalog change has to be merged by hand.

Two properties make this safe, and both are deliberate:

- **Resolved at dispatch, never in the catalog.** The capability list is published to the *retained* profile topic. Resolving there would broadcast every deployment value to anyone subscribed to the registry. On the wire the prompt stays a template; substitution happens only on the way to the executor.
- **Environment first, then arguments.** `${...}` is expanded *before* `{{args}}`, so a value arriving in an invoke argument is never itself expanded. Were the order reversed, any caller could pass `"${SOME_SECRET}"` as an argument and have the bridge read it back — turning every invoke into an environment read.

An unset variable substitutes empty and logs a warning; it never leaks the literal `${NAME}` into the executor's instructions.

Note the boundary this respects: env vars supply *identifiers and configuration*, not credentials for the mesh to use. Authentication still belongs to the executor.

Delegation: agents asking agents

Each agent owns distinct capabilities. An agent that meets work outside its own asks the agent that owns it — a human simply enters the mesh at one agent, and work flows from there.

Requests are **directed, never broadcast**. `invoke` names one agent, so a receiving agent never decides whether to serve: it was addressed for a capability it publishes, so it does the work. There is no bidding and no contention. The only choice happens on the **asking** side — which agent to ask — which is why every agent subscribes to the registry.

Discovery

Every agent subscribes to the retained registry of the whole mesh:

```
<root>/registry/+/profile    capabilities of every agent  
<root>/registry/+/status    who is online
```

Because both are retained, an agent learns the entire mesh the moment it subscribes — no announcement round, no waiting for peers to speak. An agent's own profile arrives on the wildcard and is ignored.

Requirement: the executor must have the tools

Both forms depend on the host exposing the plugin's tools to the agent. Under OpenClaw that means `tools.alsoAllow` must list `mqtt_publish`, `mesh_ask` and `mesh_peers`, because a tool profile otherwise filters plugin tools out entirely.

This fails quietly and is worth checking first: without `mqtt_publish` an executor cannot publish its result and jobs appear to hang or get re-dispatched by the watchdog; without `mesh_ask` dynamic delegation simply never occurs.

Declared delegation still works without `mesh_ask`, because the bridge performs those asks itself — which is one reason to prefer it where the dependency is known in advance.

Two ways to delegate

They fail differently, so a deployment chooses with `mesh.delegation`.

Declared — a capability lists what it depends on. The bridge performs those asks *before* the executor starts and hands it the answers:

```
{ "service": "code.review",
  "prompt": "Review PR {{pr}} in {{repo}}.",
  "delegates": [
    { "agent": "dba", "service": "schema.review", "as": "schemaReview",
      "args": { "migration": "{{migration}}" }, "required": false }
  ] }
```

Independent dependencies run concurrently. Each answer is injected under its `as` name in a `CONTEXT FROM OTHER AGENTS` block. A `required` dependency that fails fails the job before the executor starts, with a terminal result published so nobody waits; an optional one lets the job proceed and tells the executor it could not be answered.

Declared delegation is **deterministic**: it needs no tool in the executor's session and does not depend on the executor choosing to delegate. What it cannot do is adapt to what a job turns out to need.

Dynamic — the executor calls `mesh_ask` mid-job, having found the peer with `mesh_peers`. Flexible, and the only option when the need is discovered rather than known in advance. It depends on the executor deciding to call the tool, and on tools being reachable from its session.

mesh.delegation	Declared	mesh_ask
both <i>(default)</i>	✓	✓
declared	✓	✓ refused, with the reason
dynamic	✓ ignored	✓
off	✓ ignored	✓ refused

When dynamic is unavailable the executor's briefing omits the peer directory entirely, rather than advertising a tool that will refuse.

Asking

An ask is an ordinary `invoke` to the peer's command topic, with `requestedBy` set to the **asking agent's id**. Results therefore route to the asker's own owner scope, which it already subscribes to — that is the return path.

```
{ "service": "schema.review",
  "args": { "migration": "..."},
  "requestedBy": "conan",
  "jobId": "ask-mt7k2p-9f3c",
  "parentJobId": "rev-118",
  "rootJobId": "rev-118",
  "depth": 1 }
```

The asking agent **waits** for the terminal result and uses it in its own work. That is the point of delegation: conan asks dba because it needs dba's answer to finish, so the entry agent returns one complete reply rather than the requester assembling fragments.

Lineage

Every ask creates **its own job with its own id**. A chain of agents is a chain of jobs, not one job passed around, so three fields link them:

Field	Meaning
parentJobId	the job that asked for this one
rootJobId	the original request every job in the chain shares
depth	hops from the root; <code>0</code> entered the mesh directly

```
rev-118  parent –      root rev-118  depth 0
ask-9f3c parent rev-118 root rev-118  depth 1
ask-2b71 parent ask-9f3c root rev-118  depth 2
```

Without this, a five-agent chain is five unrelated ids: a failure cannot be attributed to the request that caused it, and a cancel has no way to find what to stop.

Hop limit

`mesh.maxDepth` (default `4`) bounds the chain. A request arriving deeper than the limit is **rejected before any work starts** — a cycle allowed to begin is a cycle that runs until something else stops it. The asking side refuses too, so nothing reaches the wire.

Cancel propagation

Cancelling a job cancels everything it delegated: the bridge publishes `cancel` to each peer it asked, and those peers cancel *their* children in turn. One cancel unwinds the whole chain without any agent needing to know its shape.

What this does not solve

Delegation inherits the trust model below. `requestedBy` on an ask is the asking agent's self-declared id, exactly as with a human requester — a peer cannot verify who really asked unless the broker enforces it.

Credentials & scope boundary

The bridge carries jobs, not credentials. It holds exactly two secrets: the broker username/password, and the optional `web.auth` panel token. It has no GitHub, cloud or third-party authentication of any kind, and no code path that acquires or forwards one.

Service credentials belong to the **executor**, which authenticates with the host's own configured identity (for example an already-authenticated `gh CLI`). This follows directly from "transport in the framework, logic in the agent": an agent that needs to read a private repository is exercising its own authority, not authority delegated through the mesh.

Never place tokens in `invoke` args. Payloads are readable by every subscriber to that topic, and results are **retained on the broker**, so a secret published once persists until it is explicitly overwritten.

Durability

The bridge holds one **persistent session**: `clean: false`, QoS 1 on every subscription, 30-second keepalive, 5-second auto-reconnect, LWT for presence. The connection is outbound only, so it works behind NAT.

Durability depends on the **client id being stable across restarts**. It is derived from hostname + install path — never from the process id. This matters more than it looks: with a changing id, every restart is a *new* MQTT session, so the broker's queued QoS-1 messages stay orphaned with the dead session and any `invoke` published while the gateway was down is lost silently. It also leaked one abandoned session per restart, since MQTT 3.1.1 has no session expiry.

- An `invoke` published while the bridge is down is **queued by the broker and delivered on reconnect**.
- If you genuinely run two gateways from the same install directory, set `broker.clientId` explicitly on one of them. Session durability and multi-instance safety are in tension and the operator has to choose. A collision manifests as a kick-loop, which the bridge detects and logs explicitly rather than letting it look like flaky networking.
- Set `broker.protocolVersion: 5` to get a bounded `sessionExpirySeconds` (default 24h), so a decommissioned deployment stops accumulating queued messages forever.

In-memory job tracking (`activeJobs` , `history`) does not survive a restart. Retained results replay on resubscribe, so completed jobs reappear; jobs in flight across a restart are no longer watched and rely on the executor publishing its own retained result.

Identity & trust model

MQTT delivers **topic + payload only** to subscribers. No username, no client id — the publisher's broker identity never travels with the message. Therefore:

- `owner` is **self-declared**: whatever string the sender puts in `requestedBy` .
- With the default configuration, that string is *required* but not *authenticated*. Anyone with broker credentials can claim any owner scope by forging `requestedBy` .
- Privacy/isolation is therefore **conventional, not enforced**, unless you enable verification below.

Optional owner verification (`mesh.verifyOwner`)

The bridge can verify `requestedBy` against a broker-injected `client_username` , using an EMQX rule-engine payload enrichment. When enabled it **fails closed**: an `invoke` that arrives without `client_username` is rejected, and one whose `requestedBy` disagrees with the broker identity is rejected with the mismatch reported.

This is **off by default** because it requires broker-side setup first. Wire the EMQX rule, then turn it on. Until you do, `ownerPolicy.verified` is `false` in the profile and clients should treat owner scoping as a convention.

Stronger isolation remains a broker-side concern: EMQX ACLs (`subscribe jobs/${username}/#` , `publish commands/+/invoke`) enforce it at the boundary rather than in the agent.

Owner

`owner` scopes job traffic so each requester only receives their own jobs:

- `owner` = `requestedBy` from the `invoke` payload, lowercased, restricted to `[a-z0-9_-]` (anything else → `-`), trimmed.
- Set **`requestedBy` to your stable username** (e.g. `mohanad` , `ci-github` , `kg`). It's the same value for `invoke` and for your subscription filter.

- **Omitting it is rejected.** The job is refused and a `type: "error"` result is published to `jobs/public/<jobId>/result` so the failure is never a silent drop. Previously such jobs were quietly routed to `public`, where the sender's own subscription filter would never see them.
- Set `mesh.requireOwner: false` to restore the lenient behaviour, which accepts the invoke, logs a warning, and scopes it to `public`.
- Anyone subscribing the flat `jobs/#` still sees everything (the agent's own panel does this) — clients shouldn't.

Job lifecycle topics

```
agents/jobs/<owner>/<jobId>/events    milestones: accepted, started, diff-fetched, analyzing, result
agents/jobs/<owner>/<jobId>/result    terminal payload, RETAINED, QoS 1
```

Every result carries `jobId`, `owner`, `ts`, and `type`. Known types: `review`, `already_reviewed`, `error`, `duplicate`, `cancelled`, and the generic `result`. The list is open — treat an unrecognised `type` as terminal.

These guarantees are enforced by the bridge, not by the executor. Any publish to a job topic through the `mqtt_publish` tool is normalised on the way out: `jobId`, `owner` and `ts` are injected from the topic if absent, `type` is defaulted, and **retain is forced true on result topics**. Conformance no longer depends on the executor remembering to set the flag — a forgotten `retain: true` used to mean late subscribers saw nothing at all.

Events are QoS 1 and deliberately **not** retained: a subscriber joining mid-job sees the retained result but not earlier milestones.

Job topics are **always owner-scoped**. There is no unscoped `jobs/<jobId>/...` form: a result belongs to exactly one owner and is published to exactly one topic.

Cancellation

`cancel` is **cooperative and terminal at the mesh boundary**:

1. `cancel_acknowledged` is published to the job's event stream.
2. A retained `type: "cancelled"` result is published, so listeners are never left waiting on a job that will never report.
3. Any later publish from the executor for that job is **suppressed**, keeping the client's view consistent with the acknowledgement.
4. The executor's session is dropped on a best-effort basis to stop it consuming budget.

What it does **not** guarantee is immediate termination of in-flight work. The plugin runtime exposes `run`, `waitForRun`, `getSession` and `deleteSession` — there is no abort primitive — so an executor mid-tool-call may run to completion internally. Its output is discarded rather than published. Treat `cancelled` as "no further traffic and no result will be honoured", not as "the compute stopped".

Execution & liveness

Jobs are pushed straight into an isolated executor subagent at arrival — no heartbeat, no queue, so a busy or failing main session cannot swallow a job. If the runtime has no subagent API, the bridge falls back to a system-event enqueue plus heartbeat wake, which is the only pull-shaped path that remains, and only on older runtimes.

A supervisory watchdog sweeps every 60 seconds. Its liveness signal is **whether the executor's run has settled**, not whether the executor has been chatty:

- Run still in flight → alive by definition, never re-dispatched.
- Run settled without a published result → re-dispatched, up to 2 times, then a terminal `error` result.
- Fallback path with no run handle → falls back to a 5-minute silence heuristic.
- Any job exceeding `mesh.maxJobDurationMs` (default 30 min) fails with a terminal error regardless of liveness.

The previous heuristic re-dispatched any job that worked *silently* for five minutes, which a large PR review does routinely — causing duplicate execution of exactly the most expensive jobs. Executors are still asked to publish a milestone every two minutes, but correctness no longer depends on their doing so.

What you should subscribe to

As a job sender with username `mohanad` :

```
agents/jobs/mohanad/#      ← your jobs: events + results (recommended)
agents/jobs/mohanad+/result ← only final results, no progress noise
agents/commands/conan/query/reply ← if you use query
```

Do NOT subscribe `agents/jobs/#` — that's the pre-1.1 firehose of everyone's jobs.

As the agent (conan) itself: the four `commands/conan/*` topics + `agents/jobs/#` (full history for the web panel).

Example flow

1. Publish `agents/commands/conan/invoke` :

```
{ "service": "code.review", "requestedBy": "mohanad",
  "args": { "repo": "owner/repo", "pr": 1234 } }
```

2. Listen on `agents/jobs/mohanad/#` .
3. Events arrive as `.../jobs/mohanad/<jobId>/events` ; the final review (retained) lands on `.../jobs/mohanad/<jobId>/result` .

Omit `requestedBy` and step 1 is rejected — you'll find the error on `agents/jobs/public/<jobId>/result`, not on your own scope.

Web control panel

A front end for this plugin; no MQTT in the browser. Served on `127.0.0.1:8765` by default.

Route	Purpose
<code>GET /api/profile</code>	catalog + connection state
<code>GET /api/status</code>	broker stats (uptime, rx/tx, reconnects)
<code>GET /api/jobs</code>	active jobs + recent history
<code>GET /api/events</code>	SSE stream of job/status/profile changes
<code>POST /api/invoke</code>	<code>{ service, args, requestedBy, jobId? }</code>
<code>POST /api/cancel</code>	<code>{ jobId, requestedBy }</code>
<code>POST /api/config</code>	service CRUD

The panel consumes the SSE stream and only falls back to polling if the stream drops. Set `web.auth` to require a bearer token — it is accepted as an `Authorization` header, or as `?token=` for the SSE stream, since `EventSource` cannot set headers. This setting was previously declared but never enforced; it is now enforced on every route.

Changes v1.2 → v1.3

Delegation, which is what makes this a mesh rather than a set of agents sharing a broker:

- Agents subscribe to `registry/+profile` and `registry/+status` and keep a live peer directory. Previously every agent published a catalog that nobody read.
- An agent can ask a peer and **receive the answer**. Publishing to a peer was always possible; the return path was not, so an agent could notify a peer but never use its reply.
- `parentJobId`, `rootJobId` and `depth` link the jobs of one request into a traceable chain.
- `mesh.maxDepth` bounds chain length, refusing before anything is published.
- Cancelling a job cancels what it delegated, recursively.
- Two delegation forms, selected by `mesh.delegation`: capability-declared dependencies gathered by the bridge before the executor starts, and the `mesh_ask` tool for needs discovered mid-job.

Changes v1.1 → v1.2

Protocol:

- `requestedBy` is enforced, not merely documented as required. Rejections are published rather than silently rerouted to `public`.
- Optional broker-identity verification (`mesh.verifyOwner`), fail-closed, off by default.
- `ownerPolicy` advertised in the profile and in `query` replies, so clients can read the deployment's actual enforcement instead of inferring it.
- `cancelled` added as a terminal result type; cancel now always produces a terminal result.

Conformance fixes:

- Result payload shape and the retained flag are guaranteed by the bridge rather than by executor prompt-following.

Removed:

- The v1.0 compatibility layer is gone: no unscoped `jobs/<jobId>/result` mirror, no legacy request topic, and no `topics` config block. It existed only to carry a single deployment through the 1.0 → 1.1 migration, and it forced a service name into the transport layer.
- With it goes the last hardcoded capability. The bridge no longer contains the string `code.review`, or any other service name.

Durability & delivery:

- Client id is stable across restarts, so `clean: false` actually persists the session and invokes published during downtime survive.
- Optional MQTT 5 with a bounded session expiry.
- Web panel moved from a 2.5-second poll to Server-Sent Events.
- `services.json` changes detected via `fs.watch` instead of a 30-second mtime poll.
- Watchdog liveness switched from executor chattiness to run settlement, removing duplicate execution of long silent jobs; hard per-job duration cap added.
- `web.enabled` and `web.auth` are honoured; both were previously no-ops.